# TextLing at ArGuard Shared Tasks: An Enhanced MARBERT-Based Late-Fusion Model for Multimodal Hateful Meme Detection in Arabic

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## Abstract

The classification task for ArGuard 2026 Track A, Subtask A1, identifies whether an Arabic meme is *Hateful* or *Not Hateful*. Our system combines three separately trained models. In the first experiment, we use word-level and character-level features with balanced linear classifiers. In the second experiment, we fine-tune MARBERTv2 for text classification using class-balanced focal loss. In the third experiment, we use a frozen SigLIP image encoder with balanced logistic regression. We combine the Class-1 probabilities from the three models using late fusion. On the test set, the fused model achieved a Macro-F1 score of 0.76 and an accuracy of 0.79.

## 1 Introduction

A classification task is a learning process that predicts discrete labels for data (Abiola et al., 2025a,b; Adebanji et al., 2024, 2022; Ojo et al., 2023, 2024, 2020). Classifying sentiments and emotions in multimodal content is challenging, especially in low-resource languages such as Arabic (Haouhat et al., 2025; Youcef and Barigou, 2021). Hateful memes are particularly difficult to classify because their meaning often depends on the combination of text, images, cultural context, and implicit intent (El Attar, 2025). A caption may appear harmless on its own but become hateful when paired with an image. Similarly, an image may be interpreted as hateful when combined with its accompanying text.

In Arabic, this challenge is increased by dialectal differences, non-standard spelling, code-switching, sarcasm, culturally specific references, and OCR noise. ArGuard 2026 seeks to address these challenges through a shared task on harmful Arabic content (Alam et al., 2026; Kmainasi et al., 2026). We focus on Track A, which concerns multimodal hateful-meme detection. In particular, we address Subtask A1, which requires binary classification of each meme as either *Hateful* or *Not Hateful* on the AHA-Memes benchmark, which provides fine-grained multimodal annotations for Arabic hateful memes (Kmainasi et al., 2026).

We developed a system that deliberately preserves different inductive biases instead of using a single end-to-end vision-language model. A sparse lexical branch captures short expressions, spelling patterns, and OCR-corrupted fragments; MARBERTv2 provides contextual Arabic text representations; and SigLIP supplies generic visual-semantic features. Their probabilities are fused only after independent training. Late probability fusion is performed with fixed weights of $0.10/0.30/0.60$ assigned to the lexical, MARBERTv2 and SigLIP branches, respectively. A decision threshold of $\tau = 0.50$ is subsequently applied to obtain the final label. Figure 1 summarizes the pipeline.

The main contributions are:

- A reproducible three-branch multimodal architecture that combines sparse lexical, Arabic contextual, and visual representations.
- Imbalance-aware training through balanced linear classifiers and class-balanced focal loss.
- Development-set calibration of fusion weights and the classification threshold.
- An analysis of development-set and test-set performance, including error asymmetry between hateful and non-hateful examples.

## 2 Related Work

Recent Arabic meme research views memes as multimodal content, not just text (Shah et al., 2024; El Attar, 2025; Alam et al., 2024). This means that both the written words and the image are important for understanding a meme. The authors in (Alam et al., 2024) introduced a manually annotated dataset of Arabic memes for studying propagandistic content. The authors showed that combining text and visual information improves analysis.



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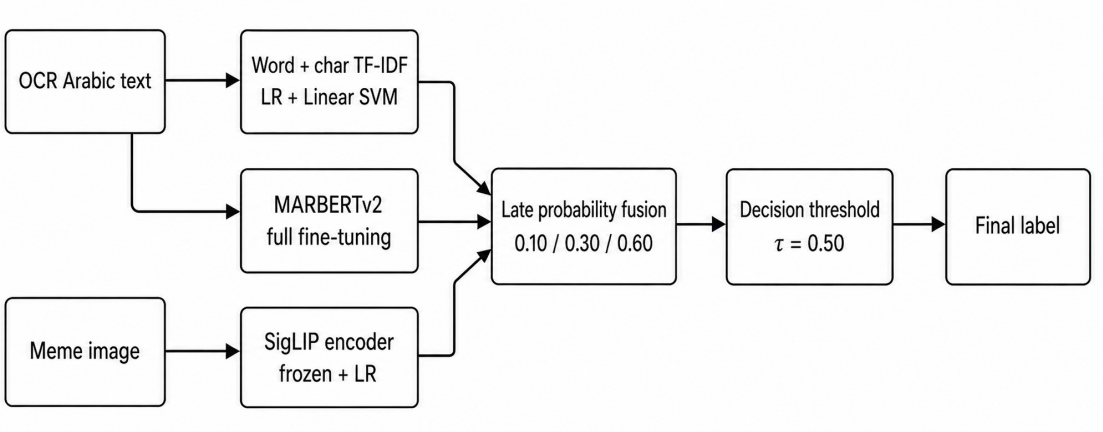



Figure 1: Architecture of the proposed late-fusion system.

The MAHED 2025 shared task also used multimodal approaches to detect hope and hate in Arabic online content (Zaghouani et al., 2025). More recently, AHA-Memes focused on fine-grained hate detection in Arabic memes by considering both image and text features (Kmainasi et al., 2026). These datasets support the need for methods that can handle noisy Arabic text while also using useful information from the accompanying images.

Traditional lexical methods remain useful for Arabic text classification, especially when the available training data are limited (Youseef et al., 2026). Word-level features can capture common words and short phrases, while character-level features can better handle spelling differences, dialectal forms, OCR errors, and repeated characters. TF-IDF representations with linear classifiers are therefore often strong baseline approaches for Arabic social-media content. These sparse features can also complement pretrained language models because they preserve direct lexical information that may not be fully captured by contextual embeddings.

Contextual information from pretrained Arabic language models provides an additional source of useful semantic features (Alammary, 2025). The contextual Arabic language models ARBERT and MARBERTv2 were developed to handle variation in Arabic, while MARBERTv2 is particularly suitable for social-media language. To represent visual and semantic information, we use SigLIP, a pretrained vision-language model that encodes image content (Zhai et al., 2023). Our system combines contextual and visual representations with sparse lexical features. This approach preserves the strengths of each model and combines their predictions at the probability level.

| Split | Not Hateful | Hateful | Total |
|---|---|---|---|
| Train | 2,176 | 1,324 | 3,500 |
| Development | 311 | 189 | 500 |
| Test | 352 | 148 | 500 |

Table 1: Dataset composition and class distribution.

## 3 Data Preprocessing Pipeline

### 3.1 Data

We used the official training, development, and test splits of the ArGuard dataset, with their sizes and class distributions summarized in Table 1. The training split was used to optimize the model parameters, while the development split supported model selection, hyperparameter tuning, and validation during experimentation. To prevent information leakage and ensure an unbiased assessment of generalization performance, the test split was excluded from all training and tuning decisions and used only for the final evaluation.

### 3.2 Text Preprocessing

We used different text preprocessing steps for the sparse lexical and MARBERTv2 branches. For the sparse lexical branch, we replaced URLs with the marker `URL`, user mentions with `USER`, and hashtag symbols with `HASHTAG`. We removed Arabic diacritics and tatweel. We also normalized several Arabic characters by mapping alif variants to a basic alif form and mapping alif maqsura to ya. For character-level features, we applied further normalization. We simplified hamza-on-waw and hamza-on-ya forms, normalized ta marbuta, and reduced repeated characters. Any sequence of

three or more repeated characters was reduced to two characters. We used word n-grams from 1 to 3 words and character-boundary n-grams from 2 to 6 characters. For MARBERTv2, we used the original OCR-extracted text without aggressive normalization. Input text was truncated to a maximum length of 160 subword tokens. Dynamic padding was applied during training and evaluation.

### 3.3 Image Normalization and Augmentation

We converted every image to RGB format and processed it with the SigLIP processor for the selected checkpoint. For each image, we extracted one embedding from the original image and another embedding from a horizontally mirrored version. We L2-normalized both embeddings, averaged them, and then normalized the final representation again. External resources were limited to pretrained model checkpoints and standard software libraries. Therefore, external knowledge in the system came only from the pretraining of MARBERTv2 and SigLIP.

## 4 System Architecture and Training Strategy

### 4.1 Sparse Lexical Ensemble

The sparse lexical branch combined word-level and character-level features. The word-level vectorizer used 1-3-grams, `min_df`=2, `max_df`=0.995, a maximum of 140,000 features, and sublinear term frequency scaling. The character-boundary vectorizer used 2-6-grams, `min_df`=2, `max_df`=0.999, a maximum of 220,000 features, and sublinear term frequency scaling. We trained two class-balanced linear classifiers: logistic regression with $C = 3.0$ and a linear SVM with $C = 0.75$. We combined their outputs as follows:

$$p_{\text{lex}} = 0.62\, p_{\text{LR}} + 0.38\, \sigma(s_{\text{SVM}}), \qquad (1)$$

where $p_{\text{LR}}$ is the positive-class probability from logistic regression, $s_{\text{SVM}}$ is the SVM decision score, and $\sigma(\cdot)$ is the sigmoid function.

### 4.2 MARBERT Text Classifier

We used `MARBERTv2` because it was pretrained on Arabic social-media text (Abdul-Mageed et al., 2021). We fine-tuned the model for binary classification and added a two-class classification head on top of the pretrained encoder. We trained MARBERTv2 using class-balanced focal cross-entropy loss (Lin et al., 2017) with $\gamma = 0.5$. We calculated class weights from inverse class frequencies raised to the power of 0.65 and then normalized the resulting weights. We used AdamW optimization with a learning rate of $2 \times 10^{-5}$ and a weight decay of 0.02. We applied cosine learning-rate decay with a 10% warmup period. The training batch size was 8, and the evaluation batch size was 32. We used gradient accumulation over 2 steps. Training was allowed to run for up to five epochs, with early stopping based on development Macro-F1 and a patience of 2 epochs. The selected configuration trained for 2 epochs. For final test inference, we retrained MARBERTv2 on the combined training and development data, for a total of 4,000 labeled examples. We used random seeds 42 and 2026. The Class-1 probabilities from the two trained models were averaged.

### 4.3 SigLIP Visual Classifier

The visual branch used the frozen `google/siglip-base-patch16-224` checkpoint (Zhai et al., 2023). The visual embedding had 768 dimensions. We kept the SigLIP encoder frozen and did not fine-tune it on the ArGuard data. We trained a class-balanced logistic regression classifier on the extracted image embeddings. The classifier used $C = 0.45$ and a maximum of 3,000 iterations. For final inference, we retrained the logistic regression classifier using the combined training and development image embeddings.

### 4.4 Late Fusion

Let $p_\ell$, $p_m$, and $p_v$ represent the positive-class probabilities produced by the lexical, MARBERTv2, and SigLIP branches, respectively. We searched for non-negative fusion weights over a coarse three-component simplex. For each candidate fusion setting, we evaluated the classification threshold on a dense grid and also included observed probability values. When several thresholds produced Macro-F1 values within 0.001 of the best score, we selected the threshold closest to 0.50. We also used a small regularization value of $2 \times 10^{-4}$ to avoid overly extreme fusion weights when several configurations performed similarly. The selected fusion setting was:

$$p_f = 0.10p_\ell + 0.30p_m + 0.60p_v, \qquad \tau = 0.50. \qquad (2)$$

| System | Macro-F1 | Acc. | Hateful F1 | $\tau$ |
|---|---|---|---|---|
| TF-IDF | 0.6842 | 0.7220 | 0.5749 | 0.5638 |
| MARBERTv2 | 0.7302 | 0.7380 | 0.6843 | 0.2872 |
| SigLIP | 0.7092 | 0.7320 | 0.6278 | 0.5425 |
| Late fusion | **0.7645** | **0.7860** | **0.6934** | 0.5000 |

Table 2: Development-set performance on 500 labeled examples.

| Evaluation | Macro-F1 | Accuracy | $\Delta$ Macro-F1 |
|---|---|---|---|
| Development | 0.7645 | 0.7860 | - |
| Test | **0.7560** | **0.7860** | -0.0085 |

Table 3: Comparison between development and test performance.

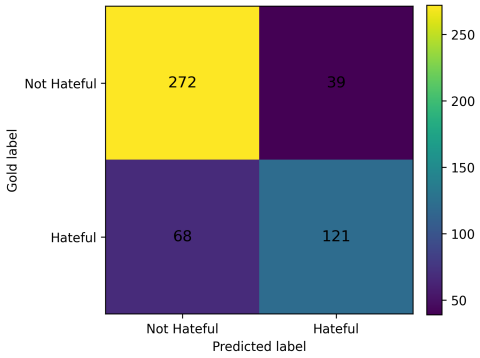



Figure 2: Development-set confusion matrix for the final late-fusion system.

## 5 Results

### 5.1 Development Results

Table 2 shows the development-set performance of the individual models and the final fusion system. The final fused system achieved a Macro-F1 score of 0.7645 and an accuracy of 0.7860. MARBERTv2 was the strongest individual model, with a Macro-F1 score of 0.7302. The final fusion improved Macro-F1 by 3.43 percentage points over MARBERTv2, 5.53 percentage points over SigLIP, and 8.03 percentage points over the sparse lexical model.

### 5.2 Test Results

Our enhanced MARBERT-based late-fusion system achieved a Macro-F1 score of **0.756** and an accuracy of **0.786** on the test set. Compared with the development results, Macro-F1 decreased by only 0.0085, while accuracy remained unchanged. This limited performance gap suggests that the system generalized well to unseen test examples and exhibited minimal overfitting. Furthermore, the superior performance of the fused system over the individual branches demonstrates that lexical, contextual, and visual representations provided complementary information for hate-speech classification.

## 6 Error Analysis

The confusion matrix in Figure 2 showed that the system correctly classified 272 *Not Hateful* memes and 121 *Hateful* memes. It produced 39 false positives and 68 false negatives. The recall for the *Hateful* class was 0.6402, while the recall for the *Not Hateful* class was 0.8746. Therefore, the most common error was classifying hateful memes as not hateful. This result was consistent with the class imbalance in the labeled data and the difficulty of detecting harmful intent when it was expressed indirectly.

## 7 Conclusion

We presented our submission to ArGuard 2026 Subtask A1. Our system combined sparse lexical features, a fully fine-tuned MARBERTv2 text classifier, and frozen SigLIP image representations through late fusion. Our results showed that combining different model types was more effective than using any single model. However, the error analysis showed lower recall for hateful content and highlighted the lack of direct image-text interaction as an important limitation.

## 8 Ethical Considerations

Harmful-content datasets include offensive, discriminatory, hateful, and sensitive material. We used the ArGuard dataset only for research purposes. We did not reproduce hateful meme examples, slurs, or unnecessary offensive content in this paper. The system processed images and OCR-extracted text for classification only and did not attempt to identify people shown in the data.

## 9 Limitations

This study is limited by the dataset size and class imbalance, which may reduce the model's ability to detect hate content and generalize to other datasets and domains. Its text branches rely on OCR, so recognition errors and token truncation may affect classification. The frozen SigLIP encoder and late probability fusion may not capture interactions between image regions and text. The fusion weights, threshold, and neural results were based on limited

development data and random seeds. Future research will examine end-to-end multimodal models with image-text interaction, OCR-aware models, and methods that preserve image information, use more random seeds and cross-domain evaluation, and apply calibration or cost-sensitive learning to improve recall for the *Hateful* class.

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